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① which optimization produced the greatest improvement?

→ The scalability enhancement produced the greatest improvement. Better thread management and efficient handling of multiple client connections reduced server crashes, improved response time and increased throughput. The application was able to support 10 concurrent clients more reliably than before.

② why is scalability important?

→ Scalability is important because it enables an application to handle an increasing number of users without significant performance degradation. A scalable system provides better resource utilization, maintains reliability under heavy load, and supports future growth with minimal changes.

③ what reliability issues did you discover?

→ The major reliability issues discovered were unexpected client disconnections, improper resource release, lack of automatic reconnection, timeout handling problems, and unhandled exceptions. These issues could interrupt communication and reduce system stability.

④ which optimization was the most difficult?

→ The most difficult optimization was implementing automatic reconnection and proper thread management. it requires a careful synchronization, exception handling and resource cleanup to prevent duplicate connections, deadlocks and memory leaks while ~~maintaining~~ maintaining uninterrupted communication.

⑤ what additional improvements are required to support 100 users?

→ To support 100 users, the application should use thread pools or asynchronous I/O, efficient load balancing, database-backed message storage, optimized memory management, connection pooling, improved logging, stronger security (authentication and encryption), and performance monitoring to ensure ~~high~~ high scalability and reliability.